

Model / Analysis Task	Purpose
1. Time Series Graph	Visualize raw stock price over time
2. Log Return Graph	Show daily returns using log differences
3. Additive Decomposition	Separate series into trend, seasonal, and residual
4. ACF of Log Returns	Assess autocorrelation in log returns
5. Stationarity - Dickey-Fuller Test	Test whether a series is stationary
6. ACF of Stock Price	Check for seasonality and autocorrelation
7. Noise Modeling	Identify random noise in the residual component
8. ARMA(1,1) Model	Model short-term autocorrelation (AR and MA terms)
9. ARMA(1,1) Residuals	Check model fit using residuals
10. ACF of ARMA(1,1) Residuals	Confirm residuals resemble white noise
11. Forecasts from ARMA(1,1)	Predict future values
12. GARCH(1,1) Model	Model time-varying volatility
13. Q-Q Plot of Log Returns	Check if log returns are normally distributed
14. ACF of Log Returns	Confirm if log returns are uncorrelated
15. Q-Q Plot of GARCH Residuals	Check normality of residuals from GARCH model
16. ACF of GARCH Residuals	Ensure residuals of volatility model are white noise
17. Asymmetric GARCH (GJR-GARCH or EGARCH)	Capture asymmetric volatility (e.g., leverage effect)
18. Q-Q Plot of Asymmetric GARCH Residuals	Same as above but for asymmetric GARCH

19. ACF of Asymmetric GARCH Residuals	Same as above
20. GARCH with t-distribution	Account for heavy tails in returns
21. Global Optimization (Global Search)	Find global minimum for likelihood-based models
22. Local Optimization (Local Search / Default)	Fine-tune model parameters locally
23. Model Implementation & Structure	Define, estimate, simulate, and evaluate time series models

Function	Purpose
<code>readtable</code>	Import Excel data
<code>datetime</code>	Convert date strings to datetime
<code>timeseries</code>	Create a time series object
<code>plot</code>	Plot time series and residuals
<code>diff, log</code>	Compute log returns
<code>seasonal</code> or custom moving average	Decompose time series (trend, seasonal, residual)
<code>autocorr, parcorr</code>	Plot ACF and PACF
<code>adftest</code>	Dickey-Fuller test for stationarity
<code>arima</code>	Define ARIMA or ARMA model
<code>estimate</code>	Fit ARIMA/GARCH models
<code>infer</code>	Compute residuals and conditional variances
<code>forecast</code>	Predict future values using a fitted model
<code>garch</code>	Define GARCH model

<code>gjr</code>	Define asymmetric GARCH model (GJR-GARCH)
<code>qqplot</code>	Generate Normal Q-Q plot
<code>GlobalSearch,</code> <code>createOptimProblem, run</code>	Conduct global optimization using multiple solvers
<code>simulate</code>	Generate synthetic time series from model
<code>resample</code>	Resample time series to consistent intervals
<code>timetable</code>	Store time series with associated timestamps

Toolbox	Used For
Econometrics Toolbox	ARIMA, GARCH, ACF, PACF, <code>estimate</code> , <code>infer</code>
Statistics and Machine Learning Toolbox	<code>qqplot</code> , model diagnostics
Global Optimization Toolbox	<code>GlobalSearch</code> , custom loss function fitting
Signal Processing Toolbox	(Optional) for additional filters or smoother